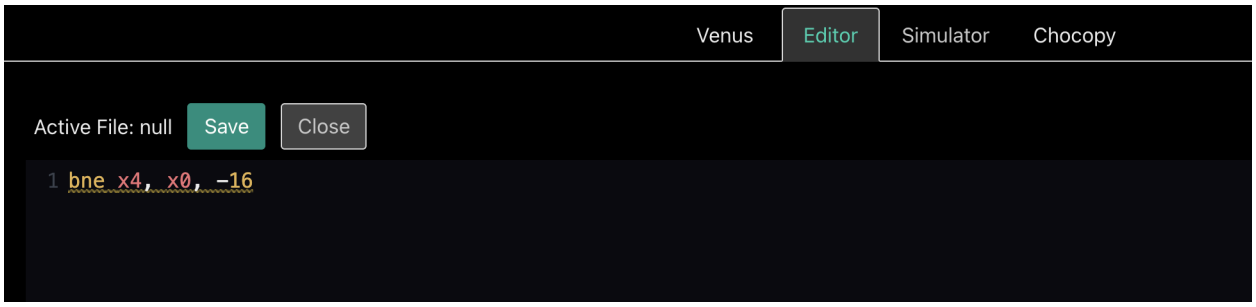
Conversion of Instructions into Machine Code

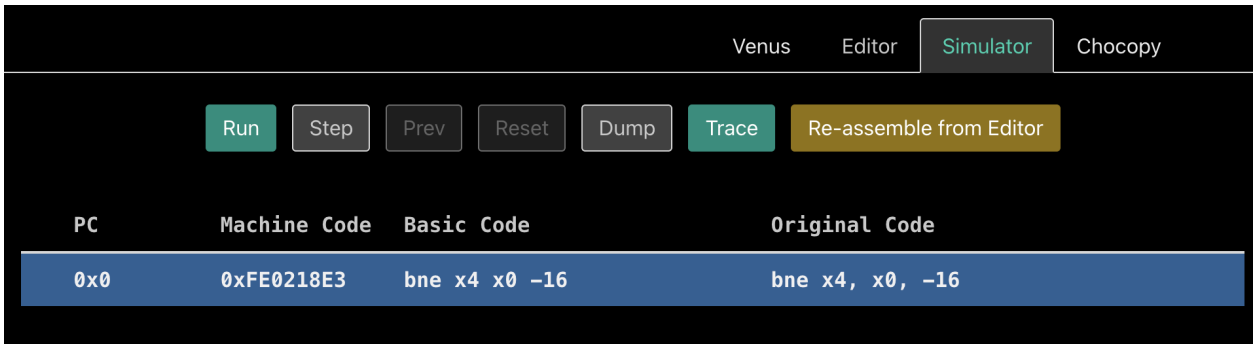
Website: https://venus.cs61c.org/

Instruction:
bne x4, x0, -16

Step 1: Write the instruction in the editor



Step 2: Go to the simulator and click on Reassemble from Editor



Conversion of Machine Code into Instruction:

Website: https://luplab.gitlab.io/rvcodecjs/#q=FE0218E3&abi=false&isa=AUTO

Machine Code: FE0218E3 and FE0218E7

Step: write the machine code and click enter

[Instruction]

FE0218E7

[Conversion]

Assembly =	jalr x17, -32(x4)
Binary =	1111 1110 0000 0010 0001 1000 1110 0111
Hexadecimal =	0xfe0218e7
Format =	I-type
Instruction set =	RV32I
Manual =	jalr

For converting C code to assembly code we have used:
https://godbolt.org/

Step 1: write your C code

C source #1

Save/Load + Add new... Vim C

```c
int main() {
    register unsigned int a asm("t1") = 94;   // x6
    register unsigned int b asm("t2") = 8;    // x7
    register unsigned int c asm("t4");        // x29

    c = (a < b);

    return 0;
}
```

Step 2: Select the compiler

RISC-V (32-bits) gcc 15.2. Compiler options...

Output... Filter... Libraries Overrides + Add new... Add tool...

```
main:
```

Step 3: The assembly code is generated:

RISC-V (32-bits) gcc 15.2.0 (Editor #1)

RISC-V (32-bits) gcc 15.2. Compiler options...

Output... Filter... Libraries Overrides + Add new... Add tool...

```
main:
        addi    sp,sp,-16
        sw      ra,12(sp)
        sw      s0,8(sp)
        addi    s0,sp,16
        li      t1,94
        li      t2,8
        mv      a4,t1
        mv      a5,t2
        sltu    a5,a4,a5
        andi    a5,a5,0xff
        mv      t4,a5
        li      a5,0
        mv      a0,a5
        lw      ra,12(sp)
        lw      s0,8(sp)
        addi    sp,sp,16
        jr      ra
```